EVIDENT-RC: Results Analysis

Evidence-checked, confidence-scored requirements classification with an open LLM

Companion report to the paper. All numbers recomputed from the per-requirement result files on 12 November 2026.

1. Summary

EVIDENT-RC asks an open language model (Llama-3.1-8B-Instruct, 4-bit weights) to label each requirement as functional (F), quality (Q) or both (F+Q), to name a quality type, and to quote the words that support each label. Code checks that every quote appears in the requirement. Five answers are sampled, and the confidence c is the share of the five that pass the check and agree with the final answer. RQ1 asks whether this finds mixed F+Q requirements more often than fine-tuned classifiers without losing overall accuracy. RQ2 asks whether c is a better confidence score than plain voting, the model's stated confidence and the probabilities of fine-tuned models.

Table 1 gives the short answer. EVIDENT-RC finds far more of the mixed requirements than any fine-tuned model on every test set: 55 of 63 on PROMISE, 50 of 81 on Dalpiaz-4 and 16 of 19 on the post-cutoff set, against at most 17, 24 and 4 for the fine-tuned models. It pays for this by calling many single-label requirements F+Q (265 F+Q answers for 63 true ones on PROMISE), so its exact match is clearly lower than RoBERTa-large on PROMISE (0.55 against 0.78). Its confidence is better calibrated than plain voting on PROMISE, but every temperature-scaled fine-tuned model gives better confidence on all three sets.

TABLE 1. HYPOTHESES AND OUTCOME

Hypothesis	Test set	Outcome
H1: higher F+Q F1 than every comparison, exact match not lower	PROMISE, 13 projects	Not supported. F+Q F1 0.34 vs 0.29 (RoBERTa-large), difference unclear; exact match lower than all three fine-tuned models (Holm $p < 0.001$).
	Dalpiaz-4	Partly. F+Q F1 0.44 beats RoBERTa-large (Holm p 0.014) and BERT-base (0.001), not clearly MiniLM or TF-IDF; exact match below RoBERTa-large (CI -0.15 to -0.01).
	Post-cutoff (82, provisional)	Partly. F+Q F1 0.53 beats RoBERTa-large (Holm p 0.002) and TF-IDF (< 0.001), not clearly BERT-base or MiniLM; no exact-match difference is clear.
H2: lower ECE and AURC than the alternatives	PROMISE	Partly. Lower ECE than plain voting and stated confidence; AURC equal to plain voting; worse than all temperature-scaled fine-tuned models.
	Dalpiaz-4, post-cutoff	Not supported against fine-tuned models; ties plain voting on Dalpiaz-4.

2. Test Sets and Statistics

Set	Items	F+Q	Role	Uncertainty
PROMISE, 13 projects	530	63	Leave-one-project-out (LOPO): each project tested once, models built on the others. P02 and P04 were used only for development.	Bootstrap over whole projects (10,000), Holm per metric
Dalpiaz-4	331	81	Four outside projects; models built on PROMISE	Bootstrap over requirements (10,000), Holm
iTrust	336	1	Functional sentences; tests false quality labels	Counts only
Post-cutoff	82	19	Public tenders from October 2024 to December 2025, after the model's training cutoff	Bootstrap over requirements (10,000), Holm

Exact match counts a requirement as right when both F and Q are right. F+Q F1 combines precision and recall for the mixed class; the false F+Q rate is the share of single-label or unlabelled requirements that a method calls F+Q. A difference is called clear only when its 95% interval excludes 0 and the Holm-adjusted p is below 0.05. The Holm families are the four baselines named in the hypothesis (and the no-quote variant on the post-cutoff set); the other rows are context.

3. RQ1: Labels

3.1 PROMISE, 13 projects

TABLE 2. PROMISE, 13 EVALUATION PROJECTS (530 REQUIREMENTS, 63 F+Q)

Method	Exact	95% CI	F+Q F1	95% CI	F+Q P	F+Q R	F+Q found / predicted	False F+Q rate	Macro-F1
EVIDENT-RC	0.55	0.49 to 0.60	0.34	0.21 to 0.44	0.21	0.87	55 / 265	0.45	0.82
L0 no quotes	0.61	0.55 to 0.65	0.36	0.24 to 0.46	0.23	0.84	53 / 229	0.38	0.83
A1 no retrieval	0.60	0.54 to 0.65	0.35	0.21 to 0.47	0.23	0.75	47 / 207	0.34	0.82
A2 no forced F+Q example	0.55	0.50 to 0.59	0.32	0.21 to 0.42	0.20	0.81	51 / 254	0.43	0.81
RoBERTa-large	0.78	0.74 to 0.82	0.29	0.14 to 0.42	0.31	0.27	17 / 55	0.08	0.88
BERT-base	0.76	0.68 to 0.81	0.20	0.08 to 0.32	0.22	0.19	12 / 54	0.09	0.86
MiniLM fine-tuned	0.74	0.69 to 0.80	0.25	0.16 to 0.33	0.29	0.22	14 / 48	0.07	0.84
Frozen MiniLM + LR	0.66	0.59 to 0.72	0.18	0.09 to 0.32	0.19	0.17	11 / 58	0.10	0.78
kNN vote (k = 8)	0.60	0.51 to 0.67	0.16	0.10 to 0.23	0.14	0.19	12 / 86	0.16	0.74
TF-IDF + SVM	0.57	0.51 to 0.65	0.14	0.06 to 0.24	0.12	0.17	11 / 96	0.18	0.73

TABLE 3. PROMISE: PAIRED DIFFERENCES (PROJECT-CLUSTERED BOOTSTRAP)

EVIDENT-RC minus	F+Q F1 difference (95% CI)	Holm p	Exact match difference (95% CI)	Holm p
L0 no quotes	-0.05 to 0.01		-0.09 to -0.03	
A1 no retrieval	-0.06 to 0.03		-0.11 to 0.00	
A2 no forced F+Q example	-0.00 to 0.03		-0.02 to 0.02	
RoBERTa-large	-0.08 to 0.15	0.387	-0.29 to -0.18	< 0.001
BERT-base	0.01 to 0.23	0.112	-0.28 to -0.13	< 0.001
MiniLM fine-tuned	-0.04 to 0.19	0.367	-0.28 to -0.12	< 0.001
Frozen MiniLM + LR	-0.03 to 0.29		-0.21 to -0.01	
kNN vote (k = 8)	0.05 to 0.28		-0.15 to 0.06	
TF-IDF + SVM	0.07 to 0.30	0.006	-0.14 to 0.07	0.599

EVIDENT-RC has the highest F+Q recall of any method (0.87), more than three times that of RoBERTa-large (0.27). Its F+Q precision is among the lowest in the table (0.21), so the F1 gain over RoBERTa-large and MiniLM is unclear, and it is clear only against TF-IDF. Exact match drops by 0.18 to 0.29 against RoBERTa-large. Even the kNN vote, which copies labels from the eight most similar training requirements, has a lower F+Q F1 (0.16), so the language model is doing more than repeating its retrieved examples.

3.2 Dalpiaz-4

TABLE 4. DALPIAZ-4 (331 REQUIREMENTS, 81 F+Q)

Method	Exact	95% CI	F+Q F1	95% CI	F+Q P	F+Q R	F+Q found / predicted	False F+Q rate	Macro-F1
EVIDENT-RC	0.61	0.56 to 0.67	0.44	0.36 to 0.52	0.35	0.62	50 / 144	0.38	0.82
L0 no quotes	0.66	0.61 to 0.71	0.47	0.38 to 0.55	0.38	0.60	49 / 129	0.32	0.83
A1 no retrieval	0.62	0.57 to 0.67	0.38	0.29 to 0.47	0.34	0.43	35 / 102	0.27	0.78
RoBERTa-large	0.69	0.65 to 0.74	0.27	0.16 to 0.37	0.42	0.20	16 / 38	0.09	0.81
BERT-base	0.57	0.51 to 0.62	0.21	0.11 to 0.30	0.29	0.16	13 / 45	0.13	0.76
MiniLM fine-tuned	0.57	0.52 to 0.63	0.32	0.23 to 0.42	0.36	0.30	24 / 67	0.17	0.76
Frozen MiniLM + LR	0.53	0.47 to 0.58	0.14	0.05 to 0.23	0.35	0.09	7 / 20	0.05	0.72
TF-IDF + SVM	0.52	0.46 to 0.57	0.32	0.22 to 0.41	0.34	0.30	24 / 70	0.18	0.72

TABLE 5. DALPIAZ-4: PAIRED DIFFERENCES (REQUIREMENT-LEVEL BOOTSTRAP)

EVIDENT-RC minus	F+Q F1 difference (95% CI)	Holm p	Exact match difference (95% CI)	Holm p
L0 no quotes	-0.08 to 0.04		-0.09 to 0.00	

EVIDENT-RC minus	F+Q F1 difference (95% CI)	Holm p	Exact match difference (95% CI)	Holm p
A1 no retrieval	-0.03 to 0.15		-0.06 to 0.05	
RoBERTa-large	0.05 to 0.30	0.014	-0.15 to -0.01	0.070
BERT-base	0.11 to 0.36	0.001	-0.03 to 0.12	0.530
MiniLM fine-tuned	0.00 to 0.24	0.082	-0.04 to 0.12	0.530
Frozen MiniLM + LR	0.19 to 0.42		0.01 to 0.16	
TF-IDF + SVM	0.01 to 0.25	0.082	0.02 to 0.17	0.051

On outside projects the fine-tuned models lose more than the language model does. RoBERTa-large falls from 0.27 to 0.20 F+Q recall, while EVIDENT-RC keeps 0.62 with better precision (0.35) than on PROMISE. Its F+Q F1 gain is clear against RoBERTa-large and BERT-base; against MiniLM and TF-IDF the intervals only just exclude 0 and the Holm p is 0.08. RoBERTa-large still has the higher exact match (0.69 against 0.61).

3.3 Post-cutoff requirements

TABLE 6. POST-CUTOFF REQUIREMENTS (82 REQUIREMENTS, 19 F+Q; PROVISIONAL LABELS)

Method	Exact	95% CI	F+Q F1	95% CI	F+Q P	F+Q R	F+Q found / predicted	False F+Q rate	Macro-F1
EVIDENT-RC	0.57	0.46 to 0.68	0.53	0.36 to 0.67	0.38	0.84	16 / 42	0.41	0.81
L0 no quotes	0.51	0.40 to 0.62	0.46	0.29 to 0.59	0.32	0.79	15 / 47	0.51	0.81
RoBERTa-large	0.63	0.52 to 0.73	0.13	0.00 to 0.30	0.18	0.10	2 / 11	0.14	0.82
BERT-base	0.63	0.52 to 0.73	0.29	0.07 to 0.50	0.44	0.21	4 / 9	0.08	0.80
MiniLM fine-tuned	0.62	0.51 to 0.72	0.21	0.00 to 0.40	0.30	0.16	3 / 10	0.11	0.80
TF-IDF + SVM	0.51	0.40 to 0.62	0.07	0.00 to 0.21	0.09	0.05	1 / 11	0.16	0.74

TABLE 7. POST-CUTOFF: PAIRED DIFFERENCES (REQUIREMENT-LEVEL BOOTSTRAP)

EVIDENT-RC minus	F+Q F1 difference (95% CI)	Holm p	Exact match difference (95% CI)	Holm p
L0 no quotes	-0.03 to 0.17	0.216	-0.02 to 0.16	1.000
RoBERTa-large	0.18 to 0.58	0.002	-0.22 to 0.09	1.000
BERT-base	-0.06 to 0.51	0.216	-0.21 to 0.09	1.000
MiniLM fine-tuned	0.06 to 0.55	0.053	-0.20 to 0.09	1.000
TF-IDF + SVM	0.25 to 0.63	< 0.001	-0.09 to 0.21	1.000

These requirements were written after the model's training data was collected, so the model cannot have memorised them. The pattern matches Dalpiaz-4: EVIDENT-RC finds 16 of 19 mixed requirements, the fine-tuned models between 1 and 4, and no exact-match difference is clear. With only 19 F+Q items the intervals are wide, and only 13 items had two annotators (Section 8).

3.4 iTrust: false quality labels

TABLE 8. ITRUST (336 SENTENCES, 311 FUNCTIONAL, 1 QUALITY)

Method	F recall	F precision	False Q labels	Exact match
EVIDENT-RC	0.94	0.95	84	0.70
RoBERTa-large	0.97	0.93	8	0.90
BERT-base	0.95	0.94	27	0.84
MiniLM fine-tuned	0.89	0.94	82	0.71
Frozen MiniLM + LR	0.91	0.95	44	0.79
TF-IDF + SVM	0.70	0.95	187	0.39

Almost every iTrust sentence is purely functional. EVIDENT-RC finds 94.2% of them but adds a quality label to 84, about ten times as many as RoBERTa-large (8). This is the same tendency seen in the PROMISE errors below, where the model adds a second label that the gold standard does not have.

4. Where EVIDENT-RC Goes Wrong

4.1 Confusion matrices

TABLE 9. EVIDENT-RC CONFUSION MATRIX, PROMISE (13 PROJECTS)

Gold \ predicted	F only	Q only	F+Q	No answer	Total
F only	129	0	69	1	199
Q only	9	108	136	3	256
F+Q	5	3	55	0	63
neither	6	1	5	0	12

TABLE 10. EVIDENT-RC CONFUSION MATRICES, DALPIAZ-4 (LEFT) AND POST-CUTOFF (RIGHT)

Gold	F only	Q only	F+Q	None		F only	Q only	F+Q	None
F only	131	0	55	1		19	0	12	0
Q only	0	22	36	1		5	12	13	0
F+Q	30	0	50	1		3	0	16	0
neither	1	0	3	0		0	1	1	0

On PROMISE the largest error is Q-only requirements called F+Q: 136 of 256. Next come F-only requirements called F+Q (69 of 199). The model rarely drops a label it should keep: only 8 of 63 mixed requirements lose one aspect. On Dalpiaz-4 the balance shifts, and 30 of 81 mixed requirements are called F only, so recall falls while precision rises.

4.2 Which quality requirements become F+Q

TABLE 11. PROMISE Q-ONLY REQUIREMENTS LABELLED F+Q, BY QUALITY TYPE

Quality type (gold)	Count	Share
Performance	34	25.0%
Security	30	22.1%
Usability	27	19.9%
Operational	27	19.9%
Maintainability	10	7.4%
Reliability	6	4.4%
no attribute	2	1.5%
Total	136	100%

The errors spread across quality types, led by performance, security, usability and operational requirements. The examples in Table 12 show the cause. The model quotes a verb such as "encoded", "run on" or "support" as the functional part. Restricting an action to some users (security) or naming a platform (operational) also makes it see a function. The annotation guideline treats several of these cases as F+Q, and PROMISE is not consistent on them, so part of this error may sit in the gold labels.

TABLE 12. EXAMPLES (PROMISE, GOLD Q ONLY, PREDICTED F+Q, WITH THE QUOTE GIVEN FOR F)

ID	Requirement	Gold type	Quoted as F
P08-417	Movies shall be encoded for 56k and 256k streams.	Operational	"encoded"
P08-428	Only administrators can activate a pre-paid card.	Security	"activate a pre-paid card"
P08-439	The website should cater to all tribes in Nigeria.	Usability	"cater to"
P14-608	The application shall run on a web application server.	Operational	"run on"
P08-425	Only registered customers can purchase streaming movies.	Security	"purchase streaming movies"
P05-203	The product must support Internet Explorer 5.5 and above.	Operational	"support"

TABLE 13. EXAMPLES OF MISSED MIXED REQUIREMENTS (GOLD F+Q)

ID	Requirement	Predicted
P06-347	The product shall have a customizable Look and Feel.	Q only

ID	Requirement	Predicted
P10-493	When the offensive player takes a shot the product shall simulate the sound of a ship at sea.	F only
P10-495	When a shot is marked as a hit the product shall simulate the sound of a shot hitting its target.	F only
P10-494	When a shot is marked as a miss the product shall simulate the sound of a shot missing its target.	F only

4.3 Per project

TABLE 14. EXACT MATCH AND F+Q FOUND PER PROMISE PROJECT: EVIDENT-RC (E) AND ROBERTA-LARGE (R)

Project	Items	Gold F+Q	Exact E	Exact R	F+Q found E	F+Q found R	False F+Q E
P01	28	7	0.57	0.64	6	0	9
P03	80	4	0.55	0.78	2	0	29
P05	73	5	0.63	0.82	5	2	23
P06	74	6	0.47	0.80	5	0	37
P07	23	2	0.56	0.74	2	1	10
P08	93	19	0.59	0.73	19	6	36
P09	24	0	0.54	0.92	0	0	11
P10	53	5	0.68	0.83	2	1	12
P11	13	3	0.54	1.00	3	3	5
P12	22	8	0.55	0.73	8	3	10
P13	19	3	0.26	0.74	2	1	12
P14	16	1	0.31	0.56	1	0	9
P15	12	0	0.42	0.92	0	0	7
All	530	63	0.55	0.78	55	17	210

RoBERTa-large has the higher exact match in 13 of 13 projects, so the gap is not caused by one project. EVIDENT-RC finds more mixed requirements in every project that has any, and finds all 19 in P08, the project with the most.

5. Ablations

TABLE 15. WHAT EACH PART ADDS (EVIDENT-RC MINUS VARIANT; 95% CI)

Variant	Exact	F+Q F1	False F+Q	Exact diff	F+Q F1 diff	Set
L0 no quotes	0.61	0.36	0.38	-0.09 to -0.03	-0.05 to 0.01	PROMISE
A1 no retrieval	0.60	0.35	0.34	-0.11 to 0.00	-0.06 to 0.03	PROMISE
A2 no forced F+Q example	0.55	0.32	0.43	-0.02 to 0.02	-0.00 to 0.03	PROMISE
L0 no quotes	0.66	0.47	0.32	-0.09 to 0.00	-0.08 to 0.04	Dalpiaz-4
A1 no retrieval	0.62	0.38	0.27	-0.06 to 0.05	-0.03 to 0.15	Dalpiaz-4

Asking for quotes costs accuracy. Without quotes (L0) the same model has a higher exact match on PROMISE (0.61 against 0.55, interval excludes 0) and fewer false F+Q answers, and its F+Q F1 is no lower. A quote for each label seems to push the model to find something for both labels. Retrieval of eight similar examples also raises the false F+Q rate, while the single forced F+Q example has no measurable effect. On Dalpiaz-4, retrieval lifts F+Q recall from 0.43 to 0.62, which is the one place where a component clearly helps find mixed requirements.

6. RQ2: Confidence

6.1 Same predictions, different confidence signals (PROMISE)

Each signal below scores the same 530 EVIDENT-RC answers, so only the ranking and calibration of the confidence differ. "Correct" means F and Q are right, and the quality type is right where the gold standard gives one. AUROC measures how well confidence separates right from wrong answers (higher is better). ECE is the average gap between confidence and actual accuracy (lower is better). AURC is the average error rate as the system answers more and more items, most confident first (lower is better).

TABLE 16. CONFIDENCE SIGNALS ON IDENTICAL PREDICTIONS, PROMISE (PAIRED BOOTSTRAP, 10,000)

Signal	Accuracy	AUROC	c minus signal	ECE	c minus signal	AURC	c minus signal
Checked-vote share c (EVIDENT-RC)	0.504	0.632		0.265		0.406	
Plain agreement (all 5 samples)	0.504	0.612	-0.00 to 0.04	0.294	-0.04 to -0.02	0.419	-0.03 to 0.00
1 minus vote entropy (checked samples)	0.504	0.628	-0.00 to 0.01	0.240	-0.01 to 0.06	0.408	-0.00 to 0.00
1 minus vote entropy (all samples)	0.504	0.602	0.00 to 0.06	0.269	-0.03 to 0.02	0.426	-0.04 to -0.00
S3: mean of c and stated confidence	0.504	0.636	-0.01 to 0.00	0.371	-0.12 to -0.09	0.404	-0.00 to 0.01

Checking the quotes lowers ECE against plain agreement by 0.02 to 0.04, a clear but small gain. It does not change the ranking: AUROC and AURC intervals include 0 against plain agreement and against vote entropy.

6.2 Whole systems

TABLE 17. END-TO-END CONFIDENCE: EACH SYSTEM WITH ITS OWN ANSWERS

System	Accuracy	AUROC	ECE	AURC	Set
EVIDENT-RC (checked vote, c)	0.504	0.632	0.265	0.406	PROMISE
S1: plain voting	0.494	0.631	0.314	0.418	PROMISE
S2: stated confidence (1 sample)	0.445	0.529	0.549	0.540	PROMISE
RoBERTa-large, temperature-scaled	0.702	0.770	0.074	0.153	PROMISE
BERT-base, temperature-scaled	0.643	0.743	0.105	0.215	PROMISE
MiniLM fine-tuned, temperature-scaled	0.626	0.795	0.092	0.197	PROMISE
EVIDENT-RC (checked vote, c)	0.613	0.632	0.169	0.304	Dalpiaz-4
S1: plain voting	0.601	0.654	0.182	0.298	Dalpiaz-4
S2: stated confidence (1 sample)	0.553	0.544	0.438	0.424	Dalpiaz-4
RoBERTa-large, temperature-scaled	0.695	0.737	0.090	0.156	Dalpiaz-4
BERT-base, temperature-scaled	0.568	0.762	0.045	0.258	Dalpiaz-4
MiniLM fine-tuned, temperature-scaled	0.574	0.748	0.054	0.261	Dalpiaz-4
EVIDENT-RC (checked vote, c)	0.476	0.668	0.263	0.401	Post-cutoff
RoBERTa-large, temperature-scaled	0.561	0.824	0.185	0.220	Post-cutoff
BERT-base, temperature-scaled	0.476	0.818	0.137	0.283	Post-cutoff
MiniLM fine-tuned, temperature-scaled	0.549	0.747	0.192	0.251	Post-cutoff

The model's stated confidence is nearly useless: it says 0.9 or more for almost every answer, so its ECE is about twice that of c. Plain voting and c rank errors equally well. Every temperature-scaled fine-tuned model has a higher AUROC, a lower ECE and a lower AURC than c on all three sets. On PROMISE all these differences are clear (Holm $p < 0.001$); on Dalpiaz-4 the AURC gap is clear only for RoBERTa-large. Part of this follows from accuracy: a model that is right more often has fewer errors to rank.

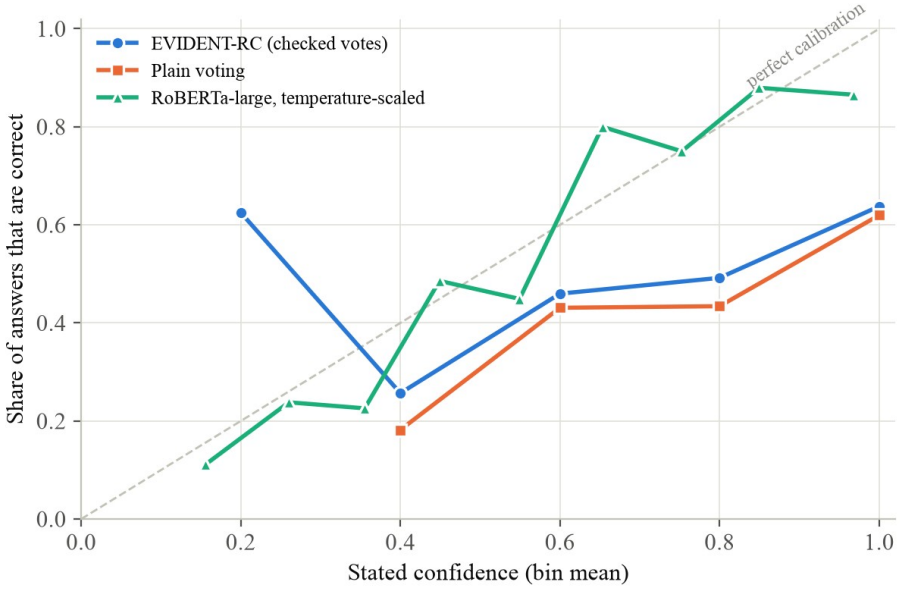


Fig. 1. Reliability on PROMISE. Points show the share of correct answers in each confidence group; the dashed line is perfect calibration. The six EVIDENT-RC groups (c = 0, 0.2, 0.4, 0.6, 0.8 and 1) hold 4, 8, 74, 124, 124 and 196 answers; groups under 5 answers are not drawn.

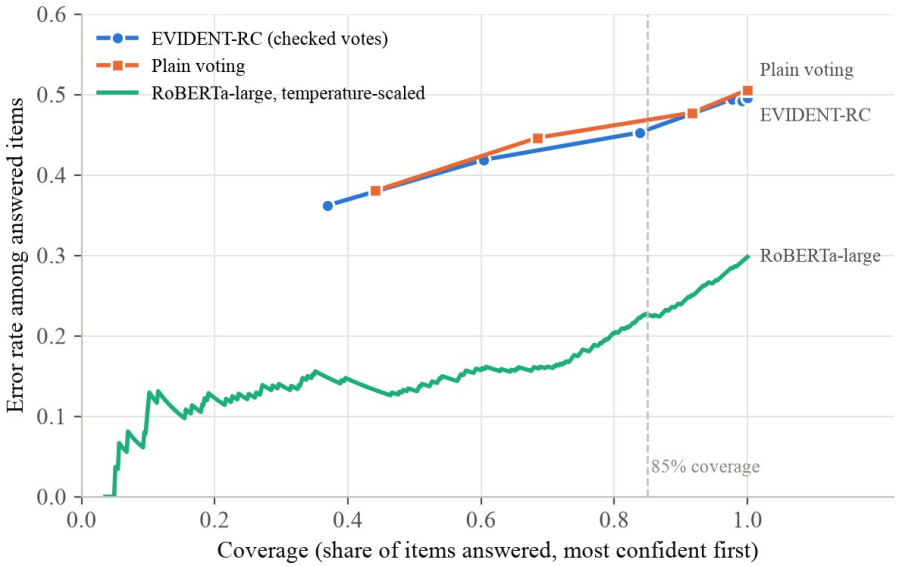


Fig. 2. Risk against coverage on PROMISE. Moving left, the least confident answers are handed to a person. Even at full confidence (c = 1) more than a third of EVIDENT-RC answers are wrong.

7. Faithfulness, Stability and Memorisation

TABLE 18. DO THE QUOTES DRIVE THE LABELS? (PROMISE, CHECKED ANSWERS)

Test	F label	Q label
Answers tested	413	368
Label dropped when the quote is masked	11.4%	15.8%
Label dropped when random words of the same length are masked	7.3%	7.1%
Label kept when only the quote is shown (sufficiency)	76.8%	82.9%
Label still given when the whole requirement is masked	65.2%	81.2%

Masking the quote removes the label more often than masking random words (paired intervals 0.007 to 0.075 for F and 0.052 to 0.125 for Q), so the quotes carry some of the decision. The effect is small, though. With the whole requirement masked the model still gives F 65.2% and Q 81.2% of the time, which means the labels rest largely on the retrieved examples and the model's prior. 91.8% of samples pass the quote check, and the checked and plain votes pick the same answer for 96.4% of requirements, so the check rarely changes a prediction.

TABLE 19. THREE SAMPLING SEEDS (PROMISE, 5 SAMPLES EACH)

Seed	Exact	F+Q F1	False F+Q	F+Q predicted	AUROC	ECE
42	0.551	0.335	0.450	265	0.632	0.265
43	0.562	0.331	0.426	251	0.641	0.241
44	0.532	0.310	0.460	266	0.667	0.275

Seed changes move exact match by at most 0.03, far less than the gap to the fine-tuned models.

TABLE 20. MEMORISATION PROBE: OVERLAP (ROUGE-L) OF THE MODEL'S COMPLETION WITH THE TRUE SECOND HALF OF EACH REQUIREMENT

Set	Items	Source named	No source	Difference	95% CI
PROMISE	605	0.165	0.153	0.012	0.00 to 0.02
Dalpiaz-4	324	0.165	0.154	0.011	-0.00 to 0.02
Post-cutoff	78	0.152	0.119	0.033	0.01 to 0.06

No completion reproduced a requirement exactly. Naming the source raises the overlap slightly on every set, and most on the post-cutoff requirements, which the model cannot have seen. The small PROMISE gap is therefore what any hint about the source produces, and gives no sign of memorisation. Without the JSON grammar, on one PROMISE fold (P01, 28 requirements), 7 of 140 samples were invalid JSON and every requirement still got an answer.

8. Post-Cutoff Annotation

101 requirements were copied verbatim from four public software tenders (October 2024 to December 2025). Each was assigned to two of three annotators with a software engineering background, who followed a written guideline with definitions and hard-case rules and first labelled 15 PROMISE practice items. Two annotators have returned labels so far; the third has not.

Measure	Value
Annotator A, practice items matching PROMISE gold (F/Q combination)	14 of 15 (quality type 9 of 10)
Annotator B, practice round	Not done
Main items labelled	82 of 101 (A only 53, B only 15, both 13, of which 1 later corrected)
Agreement on the 13 shared items, F/Q combination	9 of 13; Krippendorff's alpha 0.53 (F 0.44, Q 0.56); type 4 of 5
Final labels	F only 31, Q only 30, F+Q 19, neither 2

TABLE 21. ADJUDICATION LOG (OWNER DECISIONS BY THE GUIDELINE RULES)

Item	Requirement	A	B	Rule applied
EX-034	Ability to connect on a desktop, tablet, laptop or mobile phone	F=0 Q=1 Operational	F=0 Q=1 Maintainability	Working on several platforms: Q only, Maintainability
EX-037	The platform must integrate with financial and accounting systems to enable real-time tracking, automated reimbursements, and reconciliation.	F=1 Q=0	F=0 Q=1 Operational	Connecting to another system: Q only, Operational
EX-054	The system must integrate seamlessly with MSU’s existing Ellucian Banner system and have real-time integration options available.	F=1 Q=0	(not labelled)	Connecting to another system: Q only, Operational
EX-070	It must also support Single Sign-On (SSO) functionality and be fully compatible with MSU’s Multi-Factor Authentication (MFA) requirements to enhance security and streamline user access.	F=1 Q=1 Security	F=0 Q=1 Security	Who may access the system in general (login, SSO, MFA): Q only, Security
EX-076	Integrate with Emphasys Elite to prepopulate demographic information	F=1 Q=0	F=0 Q=1 Operational	Connecting to another system: Q only, Operational

Item	Requirement	A	B	Rule applied
EX-079	The system will need to be able to allow authorized users to monitor and approve/deny requests and offer commentary or feedback that is viewable by the requestee.	F=1 Q=0	F=1 Q=1 Security	Who may perform one specific action: F+Q, Security

All five disagreements and one correction concern the same hard cases the model also struggles with: connections to other systems, logins and access, and support for several devices. Agreement (alpha 0.53) is below the 0.67 target, and the guideline shares its hard-case rules with the prompt, which may favour the language model on this set. The post-cutoff results stay provisional until the third annotator returns labels.

9. Deviations From the Plan

- **Prompt.** Four prompt versions were tried on the development projects P02 and P04 only. Version 4 was frozen on 6 November after the main model, on those projects, labelled most Q-only items F+Q with version 3. A partial main run with version 3 was set aside unopened. P01 was used once, before the development rule, for a format-only fix.
- **Scoring correction.** On 10 November the RQ2 definition of "correct" was fixed: it had failed every quality requirement with no gold quality type. All RQ2 numbers here use the corrected rule.
- **Hardware.** Runs used a shared GPU slice with 10 GB, so the model ran with 4-bit weights through Hugging Face transformers. Ten samples were drawn per requirement for the main runs and the first five are used.
- **Settings.** Retrieval used k = 8 throughout (chosen on P02 and P04), and the review threshold was not tuned; the full risk-coverage curve is reported instead. Fine-tuned baselines used fixed settings with no tuning. Dalpiaz-4 and post-cutoff intervals resample requirements, since four or one source projects are too few to resample.
- **Not run.** A second model, the no-forced-example variant on Dalpiaz-4, conformal prediction, human gold quotes for span agreement, the audit of the 80 PROMISE F+Q labels and quality types for Dalpiaz-4.

10. What the Results Mean

A small open model that must quote its evidence is good at noticing that a requirement mixes a function with a quality, and it keeps that ability on projects and texts it has not seen, where fine-tuned models lose it. It is also too eager: it treats almost any verb as a function, so it raises many false mixed labels and loses overall accuracy. The quotes are real and checked, and they matter somewhat to the decision, but asking for them adds false F+Q labels, and the vote among checked answers gives only a small calibration gain. For a review workflow this suggests using EVIDENT-RC as a recall-oriented second reader for mixed requirements, with its quotes shown to the reviewer, and leaving the confidence score to a fine-tuned model.